Learnable Full-Dimensional Emotion Prototypes with Angular Relational Learning for Multimodal Emotion Recognition in Conversation

***Abstract*—**

I. Introduction

II. Related Work

III. Problem Definition and Proposed Method

A. Problem Definition

1) Multimodal Emotion Recognition in Conversation: Let a conversational dataset be denoted by $\mathcal{D} = \{\mathcal{C}_n\}_{n=1}^{N}$, where conversation $\mathcal{C}_n = \{u_{n,t}\}_{t=1}^{L_n}$ contains an ordered sequence of utterances. Each utterance is associated with aligned textual, acoustic, and visual observations,

$$x_{n,t} = \left(x_{n,t}^{T}, x_{n,t}^{A}, x_{n,t}^{V}\right), \qquad y_{n,t} \in \mathcal{Y} = \{1, \ldots, C\}, \quad (1)$$

where $y_{n,t}$ is its categorical emotion label. The modality set is $\mathcal{M} = \{T, A, V\}$. The objective is to learn a contextual classifier f_Θ that predicts the emotion of every valid utterance by exploiting both conversational context and complementary multimodal evidence:

$$\hat{y}_{n,t} = \arg\max_{c \in \mathcal{Y}} p_\Theta\left(c \mid x_{n,1:L_n}^{T}, x_{n,1:L_n}^{A}, x_{n,1:L_n}^{V}\right). \quad (2)$$

Padding utterances introduced by mini-batching are excluded from every utterance-level loss using a binary validity mask.

2) SDT Backbone and Learning Objective: For each valid utterance i, the SDT backbone produces a pure self-modal representation $h_i^m \in \mathbb{R}^{d_h}$ and an interaction-enhanced representation $\widetilde{h}_i^m \in \mathbb{R}^{d_h}$ for every $m \in \mathcal{M}$. A feature-wise SDT gate fuses the enhanced representations:

$$a_i^m = G\widetilde{h}_i^m, \quad (3)$$

$$[\beta_i^m]_k = \frac{\exp([a_i^m]_k)}{\sum_{q \in \mathcal{M}} \exp([a_i^q]_k)}, \quad (4)$$

$$\widetilde{h}_i = \sum_{m \in \mathcal{M}} \beta_i^m \odot \widetilde{h}_i^m, \quad (5)$$

$$p_i^{\mathrm{SDT}} = \mathrm{softmax}(W_o\widetilde{h}_i + b_o), \quad (6)$$

where $k \in \{1, \ldots, d_h\}$ indexes feature dimensions and $\odot$ denotes element-wise multiplication.

The SDT training objective combines the class-weighted loss of the fused classifier, three unimodal student losses, and self-distillation from the fused prediction to the modality students:

$$\mathcal{L}_{\mathrm{SDT}} = \gamma_1\mathcal{L}_{\mathrm{task}} + \gamma_2\mathcal{L}_{\mathrm{student}} + \gamma_3\mathcal{L}_{\mathrm{KD}}. \quad (7)$$

For a set $\mathcal{I}$ of valid utterances in a mini-batch and class weights ω_c, the first two terms are

$$\mathcal{L}_{\mathrm{task}} = -\frac{1}{Z_\mathcal{I}} \sum_{i \in \mathcal{I}} \omega_{y_i} \log p_i^{\mathrm{SDT}}[y_i], \quad (8)$$

$$\mathcal{L}_{\mathrm{student}} = -\sum_{m \in \mathcal{M}} \frac{1}{Z_\mathcal{I}} \sum_{i \in \mathcal{I}} \omega_{y_i} \log p_i^{m}[y_i], \quad (9)$$

where $Z_\mathcal{I} = \sum_{i \in \mathcal{I}} \omega_{y_i}$. The distillation term is

$$\mathcal{L}_{\mathrm{KD}} = \frac{1}{|\mathcal{I}|} \sum_{m \in \mathcal{M}} \sum_{i \in \mathcal{I}} D_{\mathrm{KL}}\left(p_i^{\mathrm{SDT},(\tau_s)} \,\|\, p_i^{m,(\tau_s)}\right), \quad (10)$$

where τ_s is the distillation temperature. The proposed method preserves the SDT encoder, enhanced-feature fusion, self-distillation, and final classifier, while introducing prototype-based geometric supervision over the pure modality representations.

B. Learnable Full-Dimensional Prototype Learning

1) Overview: The proposed framework augments SDT with a shared bank of learnable emotion prototypes in a complete D-dimensional space. Three modality-specific projectors map pure text, audio, and visual features into a Poincaré ball, where $D = 16$. Each class prototype combines an angular Emotion-Wheel reference with a learnable residual orthogonal to the reference plane. Prototype cross-entropy supplies discriminative supervision. Anchor regularization limits prototype drift, while an angular relational loss transfers the neighborhood structure of the Emotion Wheel to the learned hyperbolic space.

The prototype branch provides an auxiliary geometric learning signal, while the SDT classifier remains responsible for the final categorical prediction.

2) Poincaré Projection of Modality Features: We employ the curvature--1 Poincaré ball

$$\mathbb{D}^D = \{z \in \mathbb{R}^D : \|z\|_2 < 1\}. \quad (11)$$

For modality m, an affine map projects a pure feature into the tangent space at the origin:

$$v_i^m = W_m h_i^m + b_m, \qquad v_i^m \in T_0\mathbb{D}^D \simeq \mathbb{R}^D. \quad (12)$$

The exponential map produces the hyperbolic utterance representation

$$z_i^m = \mathrm{Exp}_0(v_i^m) = \tanh(\|v_i^m\|_2)\frac{v_i^m}{\|v_i^m\|_2} \in \mathbb{D}^D, \quad (13)$$

with the value at $v_i^m = 0$ defined by continuity. Sample radii are not fixed, allowing the model to use both angular and radial information. The geodesic distance between $x, y \in \mathbb{D}^D$ is

$$d_{\mathbb{D}}(x,y) = \operatorname{arcosh}\left(1 + 2\frac{\|x-y\|_2^2}{(1-\|x\|_2^2)(1-\|y\|_2^2)}\right). \quad (14)$$

3) Angular Emotion References: Each emotion class c is assigned an angular position θ_c on the Emotion Wheel. Its two-dimensional reference is

$$q_c = r_0 \begin{bmatrix}\cos\theta_c & \sin\theta_c\end{bmatrix}^\top \in \mathbb{R}^2, \quad (15)$$

where $r_0 > 0$ determines the reference scale in the tangent space. Rather than padding q_c with zeros, we introduce a learnable orthonormal map

$$U \in \mathbb{R}^{D\times 2}, \qquad U^\top U = I_2, \quad (16)$$

which embeds the angular reference into a two-dimensional subspace of $T_0\mathbb{D}^D$. The matrix U is learned jointly with the network subject to the orthonormality constraint. Consequently, the reference plane can orient itself according to the multimodal representations, while the angular inner products and distances among $\{q_c\}_{c=1}^C$ are preserved by the isometry. The embedded tangent-space anchor is

$$\bar{u}_c = U q_c. \quad (17)$$

4) Learnable Full-Dimensional Emotion Prototypes: For each class, we introduce a learnable vector $r_c \in \mathbb{R}^D$. Its component within the angular reference subspace is removed:

$$r_c^\perp = (I_D - UU^\top) r_c. \quad (18)$$

The resulting prototype is obtained by mapping the combined tangent vector to the Poincaré ball:

$$\boxed{p_c = \operatorname{Exp}_0\big(U q_c + \alpha r_c^\perp\big)} \in \mathbb{D}^D. \quad (19)$$

where $\alpha \geq 0$ controls the degree of data-driven adaptation. The term $\bar{u}_c$ preserves the angular Emotion-Wheel reference, while $r_c^\perp$ provides $D-2$ directions for learning discriminative class structure that is not specified by the two-dimensional angular prior. A single prototype bank $\mathcal{P} = \{p_c\}_{c=1}^C$ is shared by all three modalities, preventing independently drifting modality-specific class geometries.

The construction separates two complementary roles. Learning U changes the orientation of the angular reference plane but does not turn its two-dimensional structure into D independent directions. The residuals $\{r_c^\perp\}$ provide the additional degrees of freedom that allow the class prototypes to occupy the full ambient space. The orthogonality constraint prevents those residuals from overwriting the angular coordinates within $\operatorname{span}(U)$.

5) Anchor-Preserving Regularization: Unconstrained residuals could dominate the angular component and cause the prototypes to lose their intended interpretation. We therefore penalize their squared norm:

$$\mathcal{L}_{\text{anchor}} = \frac{1}{C}\sum_{c=1}^{C} \|r_c^\perp\|_2^2. \quad (20)$$

This term regulates, rather than eliminates, adaptation. An excessively large weight would shrink every residual toward zero and collapse the prototype bank back to an embedded two-dimensional angular prior.

6) Prototype Classification Objective: For utterance i, modality m, and class c, the prototype score is the negative Poincaré distance:

$$s_{i,c}^m = -\frac{d_{\mathbb{D}}(z_i^m, p_c)}{\tau_p}, \quad (21)$$

where $\tau_p > 0$ is the prototype temperature. Defining

$$\rho_{i,c}^m = \frac{\exp(s_{i,c}^m)}{\sum_{k=1}^C \exp(s_{i,k}^m)}, \quad (22)$$

the prototype cross-entropy is

$$\mathcal{L}_{\text{ProtoCE}} = -\frac{1}{|\mathcal{M}|\,|\mathcal{I}|}\sum_{m\in\mathcal{M}}\sum_{i\in\mathcal{I}} \log \rho_{i,y_i}^m. \quad (23)$$

This objective jointly updates the modality projectors, the relevant SDT encoder features, and the learnable prototype residuals. Sharing the prototypes across modalities also supplies a common class-level semantic reference without requiring direct equality between heterogeneous modality features.

7) Angular Relational Learning in the Full-Dimensional Space: Prototype classification alone does not specify how emotion classes should be organized relative to one another. We therefore transfer the angular geometry used by CPCC on the Emotion Wheel to the learned prototype bank. For classes $i \neq j$, the normalized shortest angular distance is

$$D_{ij}^W = \frac{\arccos(\cos(\theta_i - \theta_j))}{\pi}, \qquad D_{ij}^W \in [0,1]. \quad (24)$$

For every anchor class i, these distances define a target relational distribution over all other classes:

$$Q_{ij} = \frac{\exp(-D_{ij}^W/\tau_W)}{\sum_{k\neq i}\exp(-D_{ik}^W/\tau_W)}, \qquad j \neq i, \quad (25)$$

where $\tau_W > 0$ controls the concentration of the angular neighborhood.

The corresponding relations in the learned space are derived from the Poincaré distances between full-dimensional prototypes:

$$D_{ij}^P = d_{\mathbb{D}}(p_i, p_j), \quad (26)$$

$$P_{ij} = \frac{\exp(-D_{ij}^P/\tau_P)}{\sum_{k\neq i}\exp(-D_{ik}^P/\tau_P)}, \qquad j \neq i, \quad (27)$$

where $\tau_P > 0$ controls the concentration of the learned prototype neighborhood. Self-relations are masked in both Q and P. Their discrepancy is minimized through

$$\boxed{\mathcal{L}_{\text{relation}} = \frac{1}{C}\sum_{i=1}^{C} D_{\mathrm{KL}}(Q_{i,\neg i} \,\|\, P_{i,\neg i})}. \quad (28)$$

This objective preserves relative angular neighborhoods. It neither confines the prototypes to a two-dimensional circle nor forces hyperbolic distances to equal angular distances. Its influence is controlled by λ_r; an excessive weight can keep the learned geometry unnecessarily close to the original wheel and leave the additional residual directions underused.

8) Overall Learning Objective: The complete learning objective is

$$\boxed{\begin{aligned}\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SDT}} + \lambda_p \mathcal{L}_{\text{ProtoCE}} \\ + \lambda_r \mathcal{L}_{\text{relation}} + \lambda_a \mathcal{L}_{\text{anchor}}\end{aligned}}. \tag{29}$$

Here, λ_p, λ_r, and λ_a balance class-level prototype discrimination, angular relational preservation, and anchor-preserving regularization, respectively. The resulting objective jointly shapes modality representations and class prototypes while retaining the SDT categorical decision function

$$\hat{y}_i = \arg\max_{c \in \mathcal{Y}} p_i^{\text{SDT}}[c], \tag{30}$$

for the final emotion prediction.

9) Geometric Interpretation: The angular prior remains two-dimensional even though its embedding matrix U is learnable. The D-dimensional Poincaré ball is the learned representation space, not a collection of D independent angular axes. The plane $\text{span}(U)$ carries the original wheel references, whereas its orthogonal complement supplies class-specific directions through $r_c^{\perp}$. Consequently, the high-dimensional capacity arises from the learnable residuals rather than from rotating the reference plane alone. The angular relational objective constrains only relative neighborhoods, leaving the residual space available to encode discriminative structure supported by the multimodal data.

IV. Experimental Setup

V. Experimental Results

VI. Conclusion